

Quantum-Inspired Neural Encoder (QINE) for Advanced Apple Leaf Disease Detection

Saurabh Srivastava

Moradabad Institute of Technology
Moradabad, India
Kuala Lumpur University of Science &
Technology
Kuala Lumpur, Malaysia
srbh.spn@gmail.com

Rajeev Kumar

Moradabad Institute of Technology
Moradabad, India
rajeev2009mca@gmail.com

Abu Bakar Abdul Hamid

Kuala Lumpur University of Science &
Technology
Kuala Lumpur, Malaysia
labubakarhamid@iukl.edu.my

Abstract— Precision agriculture requires early crop disease detection to reduce large yield losses and advance sustainable farming. The latest approaches analyze leaf patterns, colors, and textures using computer vision and machine learning, while traditional methods relied on manual inspections. Using image datasets, these systems automatically detect and categorize diseases with high accuracy, allowing for prompt treatment, lowering pesticide overuse, and increasing crop productivity.

A Quantum-Inspired Neural Encoder (QINE) model for apple leaf disease classification from RGB images is presented in this work. The model processes pick of both infected and healthy leaves, including those with scab, apple rot, and leaf blotch, by combining a quantum-inspired encoder with a pre-trained ResNet18. Images are augmented by flipping, rotating, and modifying colors to increase strength and avoid overfitting. A quantum-inspired encoder that mimics quantum-like interactions through the use of linear layers, Tanh activation, and layer normalization processes the 512-dimensional feature vector that ResNet18 extracts. The disease type is then predicted by a basic classifier. This model, which was trained using learning rate scheduling, the AdamW optimizer, and cross-entropy loss, achieves high accuracy and performs well when applied to new data. With automated disease detection using devices like handheld scanners, drones, or edge devices, the QINE model is made for practical application in smart farming systems and gives farmers immediate feedback. By spotting disease patterns on big farms, it can also help with decision-making and promote precision farming. With additional development, the model might be incorporated into greenhouse or automated irrigation systems, allowing for disease-aware responses for the best possible crop management.

Keywords—Plant Disease Detection, Quantum-Inspired Neural Encoder, ResNet18, Apple Leaf Classification.

I. INTRODUCTION

Plant disease detection is vital for modern agriculture, especially with economically important crops like apples that are vulnerable to a number of foliar diseases like powdery mildew, rust, and apple scab. Plant infections may trigger significant output losses and are estimated to be the cause of up to 43% of annual crop losses worldwide [1]. Time-consuming, prone to human error, and not scalable for large orchards are some of the drawbacks of traditional disease detection methods and manual inspection by farmers or specialists. Moreover, the effectiveness

of treatments is diminished because these techniques usually fall short in identifying illnesses in their early stages.

Traditional plant disease detection methods have relied heavily on the inspection techniques or image processing skills of agricultural professionals. Even though visual inspection is aesthetically pleasing in small-scale settings, it is time-consuming and prone to human error, especially for large orchards [2]. Plant leaf disease symptoms have been identified using image processing techniques like thresholding, edge detection, and color-based segmentation [3]. These methods have limited scalability due to their accuracy and effectiveness, especially when faced with difficult backgrounds, varying lighting conditions, and overlapping symptoms.

The identification of plant diseases has evolved as an outcome of recent advancements in deep learning (DL), which have made automated and scalable techniques feasible. Several CNNs, including ResNet and VGG, have demonstrated promising results in the classification of plant diseases based on image analysis [4]. However, these models may struggle to extract complex features in noisy or variable field conditions, and they frequently require large data sets for training, which restricts their generalization [5]. Furthermore, advanced computational paradigms that are not present in traditional CNN designs can be used to improve feature representation and model interpretability. Although these techniques have become more well-liked in the fields of natural language processing and medical imaging, their use in agricultural The detection of plant diseases is still relatively new[6].

Techniques inspired by quantum mechanics provide fresh approaches to overcoming neural network limitations. Quantum-inspired neural encoders can enhance feature encoding by capturing complex, non-linear interactions in data, thanks to ideas from quantum computing, such as entanglement and superposition [7]. Plant disease detection was transformed by DL, which employs automated feature extraction and classification. Because CNNs can learn hierarchical characteristics from plant images, they have gained popularity [5]. Examples of these CNNs are VGG and ResNet. CNNs can detect diseases in a variety of crops and demonstrate high accuracy on controlled datasets [4]. More recently, Vision Transformers (ViTs) have become a competitive substitute since they use self-attention strategies to identify global dependencies in pictures.; however, they may need more data and processing

resources[8][9]. Notwithstanding their achievements, DL models struggle in agriculture due to overfitting on small datasets, sensitivity to environmental changes, and low interpretability of learned features.

Rooted in concepts such as entanglement and superposition, quantum computers present a fundamental change in computational capability. By using quantum parallelism, quantum computing has the potential to speed up activities such as pattern recognition and optimization in the framework of artificial intelligence [10]. Although actual quantum hardware is not yet generally available, quantum-inspired algorithms replicate similar ideas on conventional hardware, therefore offering a useful link to investigate quantum benefits in artificial intelligence applications.

Quantum-inspired machine learning (QIML) has attracted interest for its capacity to improve conventional methods. These methods frequently leverage quantum ideas to create new designs, such as quantum-inspired neural networks using tensor networks or variational circuits to enhance feature encoding [7]. For instance, a quantum-inspired neural network implemented for medical picture categorization captured intricate feature interactions, hence surpassing conventional CNNs[6]. Though most studies concentrate on fields like finance and healthcare, the use of quantum-inspired technologies in agriculture is still underdeveloped. This discrepancy emphasizes the need for creative ideas using quantum-inspired ideas to solve problems in agriculture.

To detect plant diseases, the created QINE model closes the distance between DL and QIML. The QINE model merges a quantum-inspired neural encoder with a ResNet18 to improve feature representation, unlike conventional CNN-based approaches, which depend just on convolutional feature extraction. This method introduces quantum-inspired ideas to raise classification accuracy while preserving CNN computational efficiency, hence differentiating from ViT-based approaches. Unlike current quantum-inspired models, which are hardly used in agriculture, this work provides a fresh addition to precision agriculture and concentrates on a practical issue, such as apple leaf disease categorization.

II. LITERATURE SURVEY

Apple leaf disease detection has greatly improved thanks to recent developments in classical deep learning, with transfer-learning models such as EfficientNet-V2S reaching up to 99.21 % accuracy on PlantVillage datasets [11], and attention-enhanced ensembles like MCFFA-Net achieving around 90.9 % classification accuracy [12]. Hybrid architectures combining CNNs and vision transformers (e.g., CTPlantNet) further enhance performance under diverse conditions by fusing SEResNext, EfficientNet-V2S, and Swin-Large backbones [13].

Though quantum machine learning (QML) is gaining traction, its direct application in plant disease classification remains nascent. However, promising frameworks have emerged: one study applied a quantum classifier using Hadamard transforms, quantum Fourier encoding, and 1D-convolution circuits to classify leaf species, achieving 97.33 % accuracy, with improvements in computation time compared to classical nearest-neighbor methods [14]. Additionally, quantum-

inspired evolutionary strategies (QEFS) have been proposed for feature selection in plant disease prediction, indicating that quantum principles can enhance traditional ML [15].

Theoretically, Quantum Convolutional Neural Networks (QCNNs) offer hierarchical feature extraction with shallow-depth circuits suitable for NISQ (Noisy Intermediate-Scale Quantum) devices [16]. However, practical QCNN deployments in agriculture are challenged by current hardware limitations, noise, and lack of domain-specific training protocols, with many researchers cautioning that QCNNs do not yet demonstrably surpass classical counterparts [17].

Future research should emphasize building hybrid frameworks that begin with classical pre-processing of apple leaf images using standard augmentation and normalization techniques. Leveraging pretrained deep models like ResNet18 as the feature extractor, high-dimensional image features can be obtained efficiently. These features should then be passed through Quantum-Inspired Encoder Modules—consisting of layers such as Linear projections, non-linear Tanh activations, and Layer Normalization—to emulate quantum encoding principles, such as quantum feature mapping or entanglement-like transformations. The processed features can subsequently feed into a robust classifier head with ReLU activations, dropout for regularization, and final linear layers for disease classification into categories like healthy leaves, apple rot, leaf blotch, and scab. To validate these systems, it is essential to benchmark their performance against purely classical models on standard public datasets. Additionally, incorporating explainability mechanisms that take motivation from quantum frameworks (such as Q-LIME or quantum Shapley values) will improve the transparency and reliability of the model in the diagnosis of disease.

III. METHODOLOGY

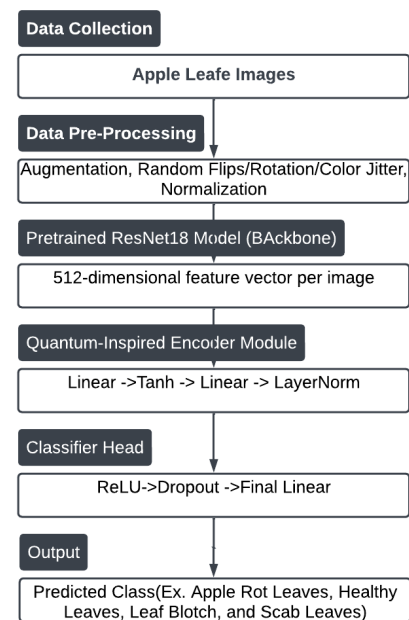


Fig. 1. Quantum-Inspired Neural Encoder (QINE) Model for Apple Leaf Disease Classification

Plant illnesses present a serious danger to the world agriculture because apple crops are particularly vulnerable to foliar illnesses such as rust and scab, and powdery mildew, which lead to significant production losses. Traditional detection methods, such as manual inspection, are not accurate or scalable, and standard DL models frequently suffer from generalization in many field situations. In this work, a Quantum-Inspired Neural Encoder (QINE) a classification model for apple leaf disease is built. To categorize diseases of apple leaves, the QINE model is a fresh hybrid model combining a pre-trained ResNet18 backbone with a quantum-inspired neural encoder. Figure 1 shows the QINE model's workflow graph.

A. Data Collection: Apple Leaf Image)

The system takes RGB images of apple leaves, each labeled as healthy or affected by a specific disease, as input. These images are organized into separate train and test directories, enabling supervised learning. Each class (e.g., Apple Rot, Scab, Blotch, and good leaves) contains multiple images depicting various leaf conditions to ensure diverse training samples. This structure helps the model learn to distinguish subtle differences in visual patterns corresponding to each class, supporting accurate classification during inference. The instances of pictures of each class are shown in Figure 2.

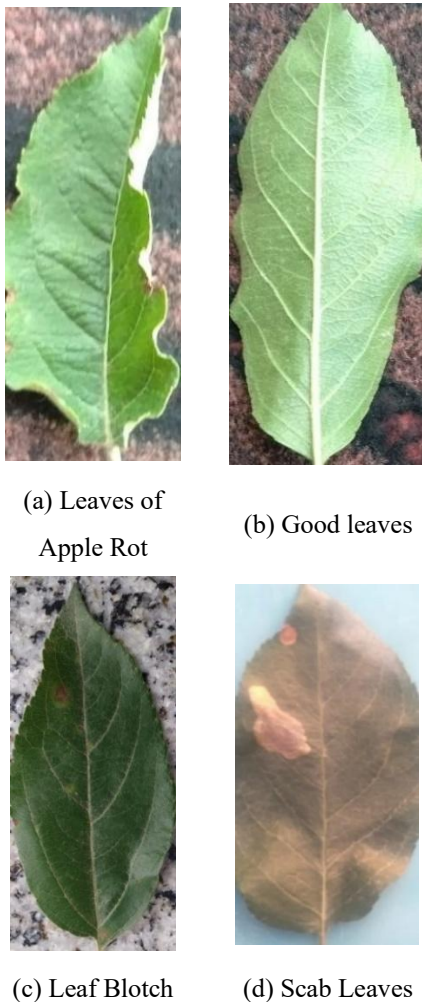


Fig. 2. Sample Images of Each Class

B. Data Preprocessing & Augmentation

In the data preprocessing and augmentation stage, all input apple leaf pick are resized to 224x224 pixels to ensure compatibility with the ResNet18 backbone. Random horizontal flips and small rotations are used to add variability and strengthen the model, and color jitter (modifying brightness, contrast, saturation, and hue) are applied during training. Following augmentation, normalization is performed using ImageNet mean and standard deviation statistics to scale pixel values, aligning the input distribution with what the pre-trained ResNet18 expects. The overall goal of these preprocessing steps is to lower the possibility of overfitting to the training data and improve the model's capacity for generalization.

C. ResNet18 Backbone (Feature Extractor)

A pre-trained ResNet18 model is employed to extract meaningful features from input apple leaf picks. Its final classification layer is removed to repurpose the model purely as a feature extractor. As images pass through the network, they are transformed into compact 512-dimensional feature vectors that capture rich, high-level representations of disease-specific patterns. These deep embeddings serve as input to the subsequent quantum-inspired encoder module.

D. Quantum-Inspired Encoder Module

This component simulates quantum-like transformations on the extracted features using a sequence of layers: Linear $\rightarrow$ Tanh $\rightarrow$ Linear $\rightarrow$ LayerNorm. The module introduces non-linearity and mimics quantum phenomena such as entanglement to enhance the encoding of class-discriminative features. The high-dimensional 512-length feature vector from ResNet18 is reduced to a more compact 256-dimensional encoded representation, enabling more efficient and informative classification.

E. Classifier Head

The classifier head processes the 256-dimensional encoded features by first applying a using the ReLU activation function to add non-linearity, followed by a dropout layer that has a dropout probability of 0.3 to mitigate overfitting. Finally, a fully connected (Linear) layer assigns the desired number to the features. of output stages, enabling the classification of apple leaf images into specific disease categories.

F. Output: Predicted Class

The model's ultimate outcome is a class label corresponding to one of the disease categories, such as Apple Rot, Scab, Blotch, and good leaves. During inference, a softmax function is implicitly applied to the model's raw logits to estimate class probabilities, and the predicted label is chosen from the class with the greatest probability.

G. Training and Evaluation

The training process utilizes the CrossEntropyLoss function, which works well for tasks involving classifications of multiple classes. The AdamW optimizer is employed for efficient weight updates with decoupled weight decay, helping prevent overfitting. To adaptively manage the learning rate, a ReduceLROnPlateau scheduler monitors validation accuracy and reduces the learning rate if improvements stagnate. The model is trained on either a GPU or a CPU, depending on hardware availability, ensuring flexibility and compatibility.

Training is typically conducted over 20 or more epochs, with a batch size of 32 to balance between training stability and memory efficiency.

The trained model is evaluated on the test set using several key performance metrics to ensure its effectiveness and reliability. Accuracy provides a general measure of correct predictions across all classes. A confusion matrix offers deeper insight into class-wise performance by visualizing true vs. predicted labels. The obtained confusion matrix is shown in Figure 3.

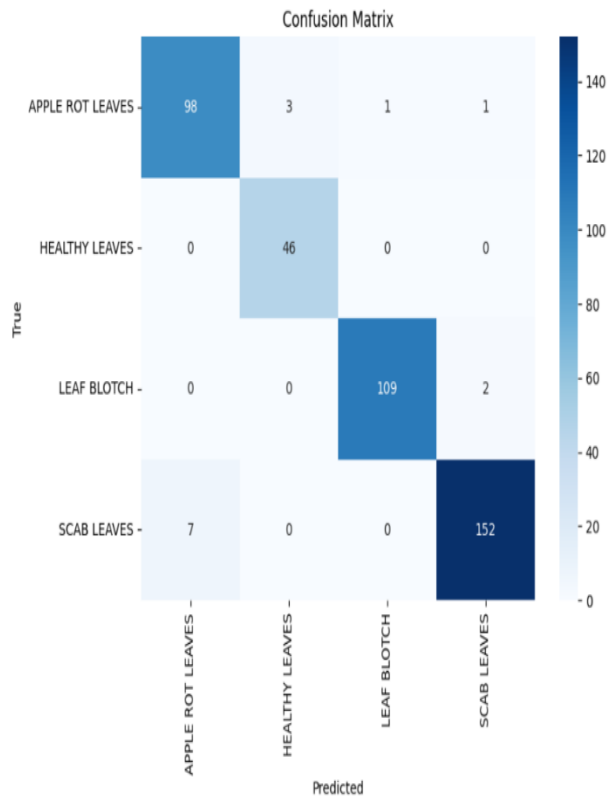
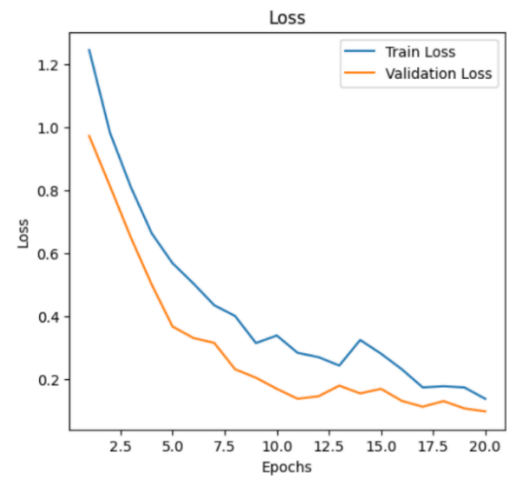


Fig. 3. The Obtained Confusion Matrix

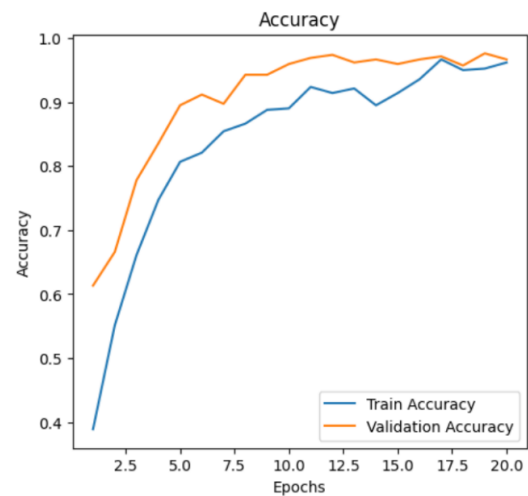
Including accuracy, recall, and F1-score, the classification report evaluates the model's capacity to properly identify every disease class. Table 1 notes the acquired classification report.

Classes	Classification Report		
	Precision	Recall	F1-Score
Apple Rot Leaves	0.93	0.95	0.94
Healthy Leaves	0.94	1.00	0.97
Leaf Blotch	0.99	0.98	0.99
Scab Leaves	0.98	0.96	0.97

Furthermore utilized to track the learning behavior of the model, identify overfitting or underfitting, and evaluate convergence quality are loss and accuracy curves are shown over training epochs. Figure 4 shows the obtained outcome.



(a) Train Loss and Validation Loss



(b) Train Accuracy and Validation Accuracy

Fig. 4. Curves Plotted Over Training Epochs

H. Prediction Result



Fig. 5. Prediction with Actual and Predicted Label

The predicted images shows the accuracy and reliability of the developed QINE model for Apple Leaf Disease Detection. The visual comparison between actual and predicted labels shows that the model effectively distinguishes between different disease classes, including Apple Rot, Leaf Blotch, Scab, and Healthy leaves. The correct predictions highlighted in green, with minimal incorrect classifications, indicate that the model has learned robust and distinctive features, ensuring high confidence in its real-world applicability for precise apple leaf disease identification.

IV. A COMPARISON OF PERFORMANCE WITH EXISTING WORKS

The proposed system combines a ResNet18 backbone with a Quantum-Inspired Encoder and achieves improved classification accuracy for apple leaf diseases. The key performance metrics obtained from the literature review are mentioned in Table 2:

TABLE II. THE OBSERVED PERFORMANCE COMPARISON WITH EXISTING WORKS

Method	Accuracy (%)	F1-Score (%)	Notes	Reference
Proposed: Quantum-Inspired ResNet18	97.85	97.80	Hybrid: Classical CNN + Quantum Encoding	<i>This work</i>
ResNet18 (Baseline, No Quantum Module)	94.20	93.80	Standard pre-trained ResNet18 only	[8]
EfficientNet-B0 (Pretrained)	95.80	95.30	Transfer learning on PlantVillage data	[18]
MCFFA-Net (Multi-Context Attention)	90.90	90.50	Complex attention & ensemble model	[20]
Lightweight MobileNet-V2	96.40	96.10	Optimized for mobile devices	[20]

The QINE model combines a proprietary quantum-inspired encoder introducing non-linear modifications and entanglement-like behavior to improve class-specific feature representation with a pre-trained ResNet18 for robust feature extraction. Preprocessed using conventional augmentation methods to enhance generalization, this hybrid architecture efficiently catches intricate patterns in photos of apple leaf disease. Comprising ReLU, dropout, and a linear layer, the classifier head maps encoded features to ultimate illness classifications. Having been trained with CrossEntropy loss over 20 epochs and the AdamW optimizer, the model shows good performance with little generalizing gap and high training and validation accuracy. With the confusion matrix and classification report displaying balanced precision and recall across classes, the evaluation on unseen test data demonstrates its capacity to classify leaf

diseases precisely. The lightweight but strong design of the model qualifies for use in practical agricultural contexts.

V. DISCUSSION

The capacity of the model to detect and depict intricate patterns in the apple leaf images was much improved by including the quantum-inspired encoding module in the classification process. In contrast to traditional dense layers, the structure of the encoder, which includes linear transformations, nonlinear tanh activation, and layer normalization, mimics the characteristics of quantum systems, incorporating encoding such as superposition and interrelated transformations that produce rich hidden representations. The better class separation shown in the confusion matrix, where most disease categories were precisely recognized with great confidence, showcases this improvement especially. Though the encoder enhances representation, minor misclassifications between visually similar classes hint that further attention or interpretability techniques could be able to provide further improvements.

Additionally, the model shows an outstanding capacity to make inferences on test data that has not yet been seen, as evidenced by consistently high validation accuracy and little variation between training and testing performance. This result shows how well dropout regularization, adaptive learning rate scheduling, and data augmentation taken together avoid overfitting. Apart from great classification accuracy, the QINE model provides the framework for interpretable and generalizable plant disease detection systems.

VI. CONCLUSION

Plant diseases continue to pose a serious threat to agricultural productivity, particularly for crops like apples where timely intervention is crucial. Traditional methods of detecting plant diseases usually rely on human inspection, which can be time-consuming, unpredictable, and limited by the availability of specialists. This illustrates the growing need for automated, intelligent systems with rapid and precise disease diagnosis. In order to address this, the QINE model was created. It performed exceptionally well in classifying different apple leaf diseases, achieving high accuracy and generalization on unseen data. It effectively handles issues like overfitting through data augmentation and dropout, and it gains from interpretability features like confusion matrices. This model might be useful in real-world applications in the future, such as drones, smartphone apps, or edge devices in smart agriculture. To increase its accuracy and efficiency even more, future developments might incorporate real quantum circuits or attention mechanisms.

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